

Lessons from autoPET: Benchmarking, Failure Analysis and Clinical Validity of Automated Lesion Segmentation in Whole-Body PET/CT

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Abstract. Quantitative PET/CT biomarkers, such as total metabolic tumor volume, predict outcomes across tumor entities, but they require whole-body lesion delineation that is too slow and reader-dependent for routine use. Deep learning has demonstrated that this delineation can be automated on curated data. What remains unclear is whether automated segmentation is robust to variation in clinical practice and whether segmentation accuracy predicts clinical usefulness. My thesis treats biomedical image analysis challenges as controlled experiments addressing both questions. Across two large MICCAI challenges (autoPET2 and autoPET3), I found that current lesion segmentation methods remain sensitive to clinically relevant distribution shifts. The dominant failure modes arise from physiological tracer uptake and underrepresented anatomical variation rather than poor lesion detection. Patient characteristics account for substantially more performance variability than differences among leading algorithms, suggesting that future progress depends on data quality, benchmark design, and uncertainty estimation. Exploring utility beyond biomarker estimation, I show that segmentation masks cannot yet be used blindly for lung cancer staging and that segmentation accuracy is a poor predictor of staging errors. A physician staging solely based on automated segmentations reached the wrong stage in a third of cases, often shifting treatment intent between curative and palliative. Automated PET lesion segmentation has moved beyond proof of concept. The next challenge is developing models that generalize in clinical practice, evaluation frameworks that reflect clinical consequences, and workflows that keep physicians in the loop.

Keywords: PET/CT · Lesion segmentation · Domain generalization · Benchmarking · Task-aware evaluation · Error propagation.

1 Research Problem and Motivation

Whole-body Positron Emission Tomography (PET) with computed tomography (PET/CT) is central to staging, response assessment, and treatment planning [5]. To date, most clinical assessments are qualitative and descriptive. Quantitative

biomarkers derived from lesion delineation, such as total metabolic tumor volume (TMTV) and total lesion glycolysis, carry prognostic and predictive value across tumor entities [22, 18]. Obtaining them requires manually segmenting lesions in whole-body PET/CT volumes. This is prohibitively slow, subject to inter-reader variability, and consequently absent from routine practice. Automated segmentation is therefore a precondition for quantitative PET/CT becoming clinically usable at scale.

Over the last decade, deep learning has largely answered the question of whether lesion segmentation for specific tumor entities can be automated. State-of-the-art methods achieve satisfactory segmentation accuracy on curated datasets and are increasingly reliable for TMTV estimation [4]. Yet these encouraging results have not translated into broadly applicable clinical tools. Current algorithms remain insufficiently understood under the types of variation encountered in practice, including differences in imaging centers, patient populations, and disease characteristics. An additional challenge arises from PET tracers, whose physiological uptake patterns differ fundamentally, like ^{18}F -fluorodeoxyglucose (FDG) and prostate-specific membrane antigen (PSMA). As novel target-specific tracers continue to emerge, repeatedly developing segmentation methods from scratch is neither practical nor scalable. Understanding how segmentation algorithms generalize across tracers is, therefore, a valuable undertaking. Finally, it remains unclear whether conventional segmentation metrics adequately reflect performance on downstream clinical tasks such as cancer staging.

This thesis is built around the hypothesis that the next major challenge in automated PET lesion segmentation is no longer improving segmentation accuracy on individual datasets, but understanding and enabling generalization. Progress, therefore, requires systematically studying how segmentation algorithms behave under clinically interesting distribution shifts, identifying the factors that fundamentally limit their robustness, and evaluating their performance in terms of downstream clinical utility rather than segmentation accuracy alone.

To investigate this hypothesis, I use biomedical image analysis challenges as scientific instruments, which align perfectly with MICCAI’s priorities. Properly designed challenges provide large-scale, reproducible experiments in which many independent solutions are evaluated under identical conditions. This makes them uniquely suited to studying algorithmic generalization, analyzing failure modes, and quantifying sources of variability at the level of an entire research community. Through the design and analysis of the autoPET2 [6] and autoPET3 [7] challenges, combined with downstream evaluation for automated lung cancer staging [11], this thesis (Fig. 1) addresses three research questions:

1. How well do current lesion segmentation algorithms generalize across clinically relevant distribution shifts, including centers, patient populations, and PET tracers?
2. Which factors fundamentally limit this generalization, and how should they be studied and benchmarked through rigorous experimental design?
3. To what extent does segmentation performance translate into other clinical tasks like staging?

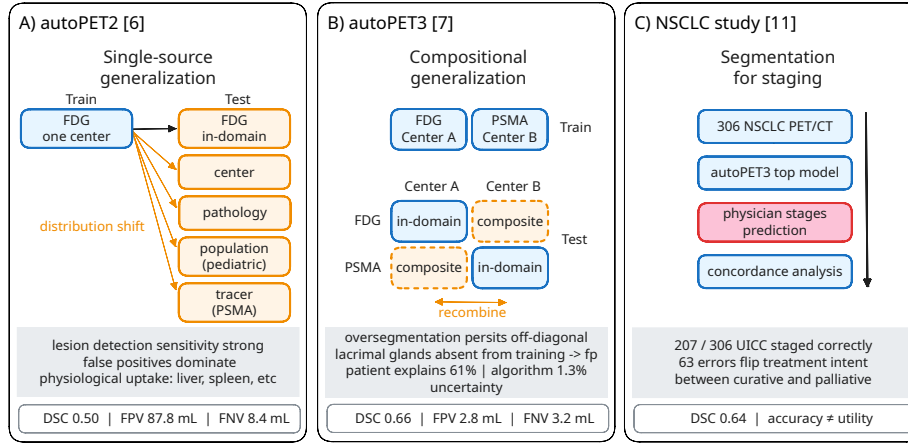


Fig. 1. Overview of this doctoral research

2 Relevance and Related Work

The last decade has seen rapid progress in automated PET/CT lesion segmentation. Early work focused on single disease entities and individual institutions, where deep learning methods demonstrated that metabolically active lesions can be segmented with satisfactory accuracy. Methodologically, there has been a shift from classification of Standardized Uptake Value (SUV)-thresholded candidate regions [23] to end-to-end detection [24] and mostly U-Net based segmentation [20, 27, 3]. With the rise of PSMA PET/CT, research teams repeated the same pattern and again developed mostly U-Net based methods [14, 12]. Few works have used unsupervised pretraining [26] or multi-tracer training with weak labels [15]. Although reported segmentation performance has steadily improved, these studies remain difficult to compare because they use different patient cohorts, evaluation protocols, and often lack external validation. Consequently, it remains unclear whether algorithmic improvements translate to more robust clinical performance.

Biomedical image analysis challenges have therefore become an important mechanism for objectively measuring progress. The first MICCAI PET segmentation challenge [10] established the value of benchmarking, yet remained limited in scale and clinical representativeness. The Head and Neck Tumor (HECKTOR) challenge series [19, 1, 2] expanded this paradigm to multi-center head-and-neck cancer segmentation and outcome prediction. More recently, the first autoPET challenge extended benchmarking to whole-body PET/CT and multiple tumor entities, demonstrating that modern segmentation algorithms perform well on large in-domain datasets but experience a marked performance loss under external distribution shifts.

3 Scientific Approach and Contributions

Biomedical image analysis challenges constitute large-scale, controlled experiments in which numerous independent solutions are evaluated on the same task under standardized conditions. This provides a unique opportunity to investigate not only which methods perform best, but also why they succeed or fail, how robust observed performance differences are, and which characteristics of the data fundamentally limit current algorithms. This thesis follows established recommendations for challenge design and reporting developed within the MICCAI community [16]. All evaluations are performed on held-out test data using containerized models in a fixed computational environment to ensure reproducibility. Statistical analyses follow established challenge methodology [25] and include ranking robustness with respect to data variability and ranking methods, metric simulations, and mixed-effects modeling to disentangle algorithmic variability from patient heterogeneity.

AutoPET2 [6] evaluates algorithms trained on a single-source FDG dataset across multiple target domains that differ in imaging center, patient population, pathology, and tracer, thereby quantifying the aggregate effect of clinically relevant distribution shifts. Building on these findings, autoPET3 [7] adopts a compositional benchmark design in which the imaging center and the PET tracer are partially crossed. By withholding specific center-tracer combinations during training, the benchmark isolates whether models can recombine previously observed sources of variation rather than merely interpolate within familiar domains. In addition, a dedicated data-centric track fixes the model architecture while allowing modifications only to the data pipeline, separating the influence of data curation and preprocessing from architectural innovation.

Across both challenges, performance is assessed using three complementary metrics, consistent with the first autoPET challenge [9]. We use the average Dice Similarity Score (DSC) as segmentation metric and false positive volume (FPV) and false negative volume (FNV) as soft detection metrics. FPV and FNV measure the volume of connected components that do not overlap with any ground truth or prediction, respectively. Beyond aggregate performance measures, extensive error analyses investigate systematic failure modes, including physiological uptake, lesion size, tracer-specific characteristics, and anatomical variability.

Finally, the thesis extends benchmark evaluation to downstream clinical decision making [11]. In this retrospective study, the highest-performing segmentation model from autoPET3 is applied to a cohort of 306 treatment-naïve non-small cell lung cancer (NSCLC) patients, where a physician performs tumor staging based solely on automated segmentations. This enables evaluation of how segmentation errors propagate into clinically relevant decisions and examines the relationship between conventional segmentation metrics and diagnostic utility.

Contributions. The thesis rests on collaborative work, and the following states what I contributed.

- **autoPET2 evaluation and reporting** [6], first author. I neither collected the data nor organized the challenge. I designed and performed the evaluation across the five target domains, the failure-mode analysis, and I wrote the study.
- **autoPET3 conception, organization, and analysis** [7], first author, under review. I co-conceived the challenge, designed the compositional evaluation and the data-centric track and all experiments, administered it, and implemented and performed the evaluation of 27 algorithms from 17 teams.
- **Downstream clinical validation in NSCLC staging** [11], second author. I co-conceived and co-designed the study, contributed to methodology, data curation, formal analysis, and investigation, and co-wrote the original draft.
- **Public resources and continuation.** I co-authored the release of the largest annotated whole-body PSMA PET/CT dataset to date [13] and co-organized the next autoPET edition on interactive segmentation [17]. I also successfully competed in the HECKTOR25 (Task 3, 5th place) and the DEEP-PSMA (3rd place) challenges.

4 Results

Generalization across domains The first question concerns whether current segmentation algorithms generalize beyond the datasets on which they were developed. AutoPET2 showed that state-of-the-art methods generalize well to external cohorts acquired with the same tracer but experience a substantial loss in performance under larger distribution shifts, particularly for pediatric and PSMA PET/CT. Across the five target domains, the best-performing algorithm achieved an average DSC of 0.50, an FNV of 8.42 mL, and an FPV of 87.84 mL. Surprisingly, most algorithms retained high sensitivity for lesion detection. This reveals that the main failure mode is false positives due to physiological uptake (e.g., in the bladder, liver, spleen, abdomen, and salivary glands).

AutoPET3 addressed this observation by shifting the single-source generalization setting to a compositional one. For that, we introduced the largest available PSMA dataset to date [13] for training. The top-ranked algorithm achieved an average DSC of 0.66, FNV of 3.18 mL, and FPV of 2.78 mL across all four test conditions. Interestingly, global multi-tracer models and tracer-specific models achieved comparable performance, suggesting that the benefits of joint modeling may be smaller than commonly assumed. Algorithms were now able to segment both tracers with satisfactory performance in the in-domain. Nevertheless, systematic errors persisted in previously unseen tracer-center combinations, primarily due to volume oversegmentation, demonstrating that even small domain changes remain challenging. The data-centric track received great feedback but remained underexplored due to the small number of participants.

Understanding what limits generalization Beyond absolute performance, the challenge studies provide insight into the factors that govern generalization. Across both benchmarks, the most persistent errors occurred in small or

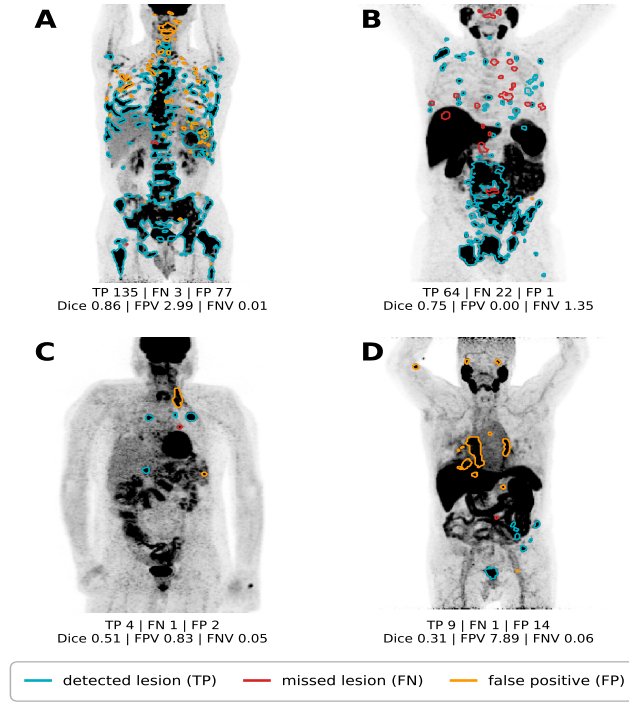


Fig. 2. Representative predictions and error types. **A** Most false positives cluster in the head and neck region, where they correspond to lesions missing from the reference annotation rather than to true model errors. **B** Small and low-uptake lesions are missed despite an otherwise high Dice score. **C** False positives follow physiological muscle uptake. **D** FPs arise at injection sites, in the lacrimal glands, and in atelectatic lung.

low-uptake lesions and in physiological structures that were underrepresented or absent during training. One illustrative example was the consistent segmentation of the lacrimal glands in the compositional PSMA cohort, a structure that was largely absent from the training data due to anonymization. Simply excluding this anatomical region recovered up to 11.5% DSC for individual algorithms, illustrating how seemingly minor dataset design decisions can substantially influence benchmark outcomes. At the same time, many remaining false positives were clinically obvious, such as tracer contamination at injection sites or physiological uptake, and could be dismissed by a physician at a glance. Conversely, qualitative inspection revealed cases in which leading algorithms correctly identified lesions missing from the reference annotations, highlighting that benchmark performance is ultimately bounded by the quality of the ground truth (Fig. 2).

The statistical analyses further showed that performance differences between leading methods are remarkably small. Bootstrap ranking analyses revealed extensive overlap among the highest-ranked submissions, while mixed-effects modeling attributed approximately 61% of unexplained DSC variance to patient het-

erogeneity but only 1.3% to differences between algorithms. In practical terms, this indicates that the patient’s characteristics have a far greater influence on segmentation performance than the choice of the best algorithm. These findings suggest that future progress is likely to come less from incremental architectural improvements and more from flagging difficult cases, improving training data, and understanding prediction failures.

From segmentation accuracy to more clinical utility The autoPET3 post-analysis exposes a fundamental limitation of current evaluation. The composite challenge score rewards volumetric agreement and therefore implicitly weights lesions by their size. Consequently, a small distant metastasis with major clinical consequences contributes little to the overall ranking. This is not a flaw of the evaluation metrics themselves, but rather reflects our limited understanding of how segmentation errors translate into clinical decision-making. What is missing is a principled way of weighting lesions according to their clinical importance.

This becomes apparent when moving beyond biomarker estimation and utilizing segmentation models for staging. Applied to 306 treatment-naïve NSCLC studies, the top-ranked autoPET3 algorithm achieved a mean DSC of 0.64. When cases are re-staged based on the artificial prediction, we correctly identified the UICC stage in 207 of 306 patients. While this demonstrates considerable potential for downstream clinical applications, 63 of the staging errors would have resulted in a significant change in treatment (curative vs palliative), i.e., real consequences for patients. Staging errors are disconnected from the per-patient DSC.

5 Open Challenges and Future Work

From autonomous to expert-supervised segmentation A central finding of this thesis is that case difficulty dominates performance far more than the choice of algorithm does. This indicates that the highest-value target is not only better fully automatic models but also an efficient human-in-the-loop workflow. Our error analysis supports this from the other side as well: a large share of the remaining false positives are ones a physician would reject at a glance, and a model struggles to rule out, such as tracer contamination at injection sites, muscle activation, or uptake in glands and the bladder. So the question shifts from how to automatically remove the last errors to how to allocate the reader’s attention well. Three directions follow: uncertainty maps that direct attention to the parts of a prediction worth checking, interactive correction that fixes an outlier case with minimal input, and triage that routes hard cases to expert review while leaving the majority, which already look excellent, alone. The agenda for the next autoPET challenges has therefore shifted towards interactive segmentation [17].

Scaling data along the right axes Expanding and releasing more data remains essential, but our findings suggest that the diversity of the data mat-

ters more than its quantity. A recent preprint by colleagues [21], to which we contributed data, reaches a similar conclusion. They trained on a large multi-institutional dataset spanning 20 disease types, ensuring spatial tumor diversity, and concluded that variability in the training dataset drives generalization. This suggests that future datasets should prioritize sources of variation that models cannot infer from existing data, including rare tumor entities and new PET tracers such as DOTATATE and FAPI, whose physiological uptake differs fundamentally from FDG and PSMA. Longitudinal imaging, treatment outcomes, and clinical context would further shift benchmark datasets from evaluating segmentation in isolation to evaluating its role in downstream clinical decision-making. Likewise, multi-reader reference standards would help distinguish algorithmic errors from inherent annotation uncertainty while enabling more reliable evaluation.

Beyond TMTV Estimation: Segmentation for Staging Tumor staging requires translating segmentation and detection performance into clinical task performance, which demands new evaluation schemes. A single missed metastasis can shift the patient’s stage and, with it, the treatment pathway between curative and palliative. Our NSCLC study confirms that model performance and staging correctness are not coupled. What is missing is knowledge of how much weight a single lesion carries in a given clinical context, which depends on its location and on the distribution of the other lesions. In ongoing work, we aim to automate the full NSCLC staging procedure using only lesion and organ segmentation combined with rule-based logic, to quantify how far current models carry this task and where the remaining gaps lie.

6 Research Vision

The central theme of this thesis is that the limiting factor in automated PET/CT image analysis is gradually shifting from building methods to understanding them. The next major challenge will be to translate both into meaningful clinical efficacy [8]. I see PET/CT lesion segmentation as a tool with significant unrealized potential. New PET tracers will provide novel insights into biological processes, while quantitative biomarkers will change how patients are triaged and treated and how new drugs are developed. Automation will also reduce variability in routine clinical workflows. From a computer vision perspective, there is often a tension between developing methods that perform well on small, closed datasets and methods that generalize across the diversity of clinical data. I believe that developing robust, generalizable methods will offer the greatest potential for downstream applications. Beyond this, open-sourcing data and models, together with closer collaboration between the computer vision and medical communities, will continue to drive innovation and help translate technical advances into clinical impact. Contributing to this transition by developing methods that bridge technical innovation and clinical application is the direction in which I hope to continue my research.

References

1. Andrearczyk, V., Oreiller, V., Abobakr, M., et al.: Overview of the HECKTOR Challenge at MICCAI 2022: Automatic Head and Neck Tumor Segmentation and Outcome Prediction in PET/CT. In: Head and Neck Tumor Segmentation and Outcome Prediction. pp. 1–30 (2023). https://doi.org/10.1007/978-3-031-27420-6_1
2. Andrearczyk, V., Oreiller, V., Boughdad, S., et al.: Automatic Head and Neck Tumor segmentation and outcome prediction relying on FDG-PET/CT images: Findings from the second edition of the HECKTOR challenge. *Medical Image Analysis* **90**, 102972 (2023). <https://doi.org/10.1016/j.media.2023.102972>
3. Blanc-Durand, P., Jégou, S., Kanoun, S., et al.: Fully automatic segmentation of diffuse large B cell lymphoma lesions on 3D FDG-PET/CT for total metabolic tumour volume prediction using a convolutional neural network. *Eur J Nucl Med Mol Imaging* **48**(5), 1362–1370 (2021). <https://doi.org/10.1007/s00259-020-05080-7>
4. Boellaard, R., Buvat, I., Nioche, C., et al.: International Benchmark for Total Metabolic Tumor Volume Measurement in Baseline 18F-FDG PET/CT of Lymphoma Patients: A Milestone Toward Clinical Implementation. *Journal of Nuclear Medicine* **65**(9), 1343–1348 (2024). <https://doi.org/10.2967/jnumed.124.267789>
5. Boellaard, R., Herrmann, K., Barrington, S.F., et al.: [18F]FDG PET/CT: EANM procedure guidelines for tumour imaging: Version 3.0. *The EANM Journal* **1**, 100006 (2025). <https://doi.org/10.1016/j.eanmj.2025.100006>
6. Dexl, J., Gatidis, S., Früh, M., et al.: AutoPET Challenge on Fully Automated Lesion Segmentation in Oncologic PET/CT Imaging, Part 2: Domain Generalization. *Journal of Nuclear Medicine* (2025). <https://doi.org/10.2967/jnumed.125.270260>
7. Dexl, J., Jeblick, K., Mittermeier, A., et al.: The autoPET3 Challenge: Automated Lesion Segmentation in Whole-Body PET/CT –Multitracer Multicenter Generalization (2026). <https://doi.org/10.48550/arXiv.2605.05775>
8. Fryback, D.G., Thornbury, J.R.: The Efficacy of Diagnostic Imaging. *Medical Decision Making* **11**(2), 88–94 (Jun 1991)
9. Gatidis, S., Früh, M., Fabritius, M.P., et al.: Results from the autoPET challenge on fully automated lesion segmentation in oncologic PET/CT imaging. *Nat Mach Intell* pp. 1–10 (2024). <https://doi.org/10.1038/s42256-024-00912-9>
10. Hatt, M., Laurent, B., Ouahabi, A., et al.: The first MICCAI challenge on PET tumor segmentation. *Medical Image Analysis* **44**, 177–195 (2018). <https://doi.org/10.1016/j.media.2017.12.007>
11. Heimer, M.M., Dexl, J., Ta, J., et al.: Artificial intelligence for TNM staging in NSCLC: A critical appraisal of segmentation utility in [18F]FDG PET/CT. *Eur J Nucl Med Mol Imaging* **53**(5), 3117–3127 (2026). <https://doi.org/10.1007/s00259-025-07677-2>
12. Jafari, E., Zarei, A., Dadgar, H., et al.: A convolutional neural network–based system for fully automatic segmentation of whole-body [68Ga]Ga-PSMA PET images in prostate cancer. *Eur J Nucl Med Mol Imaging* **51**(5), 1476–1487 (2024). <https://doi.org/10.1007/s00259-023-06555-z>
13. Jeblick, K., Schachtner, B., Mittermeier, A., Dexl, J., et al.: A Whole-Body PSMA-PET/CT dataset with manually annotated tumor lesions. *Sci Data* **13**(1), 1023 (2026). <https://doi.org/10.1038/s41597-026-07821-z>
14. Kendrick, J., Francis, R.J., Hassan, G.M., et al.: Fully automatic prognostic biomarker extraction from metastatic prostate lesion segmentations in whole-body

- [68Ga]Ga-PSMA-11 PET/CT images. *Eur J Nucl Med Mol Imaging* **50**(1), 67–79 (2022). <https://doi.org/10.1007/s00259-022-05927-1>
15. Leung, K.H., Rowe, S.P., Sadaghiani, M.S., et al.: Deep Semisupervised Transfer Learning for Fully Automated Whole-Body Tumor Quantification and Prognosis of Cancer on PET/CT. *Journal of Nuclear Medicine* **65**(4), 643–650 (2024). <https://doi.org/10.2967/jnumed.123.267048>
 16. Maier-Hein, L., Reinke, A., Kozubek, M., et al.: BIAS: Transparent reporting of biomedical image analysis challenges. *Medical Image Analysis* **66**, 101796 (2020). <https://doi.org/10.1016/j.media.2020.101796>
 17. Megne Choudja, P.O., Dextl, J., Marinov, Z., et al.: From Automation to Collaboration: The autoPET/CT-IV Challenge on Interactive Lesion Segmentation in Whole-Body PET/CT and Longitudinal CT (2026). <https://doi.org/10.2139/ssrn.6841479>
 18. Meignan, M., Cottreau, A.S., Specht, L., Mikhaeel, N.G.: Total tumor burden in lymphoma – an evolving strong prognostic parameter. *Br J Radiol* **94**(1127), 20210448 (2021). <https://doi.org/10.1259/bjr.20210448>
 19. Oreiller, V., Andrearczyk, V., Jreige, M., et al.: Head and neck tumor segmentation in PET/CT: The HECKTOR challenge. *Medical Image Analysis* **77**, 102336 (2022). <https://doi.org/10.1016/j.media.2021.102336>
 20. Park, J., Kang, S.K., Hwang, D., et al.: Automatic Lung Cancer Segmentation in [18F]FDG PET/CT Using a Two-Stage Deep Learning Approach. *Nucl Med Mol Imaging* **57**(2), 86–93 (2023). <https://doi.org/10.1007/s13139-022-00745-7>
 21. Pires, M., Gutschmayer, S., Bergalla, E., et al.: Fully automated segmentation of [18F]FDG- and PSMA-PET/CT images via data-centric Deep-Learning. (2026). <https://doi.org/10.21203/rs.3.rs-9407530/v2>
 22. Sasanelli, M., Meignan, M., Haioun, C., et al.: Pretherapy metabolic tumour volume is an independent predictor of outcome in patients with diffuse large B-cell lymphoma. *Eur J Nucl Med Mol Imaging* **41**(11), 2017–2022 (2014). <https://doi.org/10.1007/s00259-014-2822-7>
 23. Sibille, L., Seifert, R., Avramovic, N., et al.: 18F-FDG PET/CT Uptake Classification in Lymphoma and Lung Cancer by Using Deep Convolutional Neural Networks. *Radiology* **294**(2), 445–452 (2020). <https://doi.org/10.1148/radiol.2019191114>
 24. Weikert, T., Jaeger, P.F., Yang, S., et al.: Automated lung cancer assessment on 18F-PET/CT using Retina U-Net and anatomical region segmentation. *Eur Radiol* **33**(6), 4270–4279 (2023). <https://doi.org/10.1007/s00330-022-09332-y>
 25. Wiesenfarth, M., Reinke, A., Landman, B.A., et al.: Methods and open-source toolkit for analyzing and visualizing challenge results. *Sci Rep* **11**(1), 2369 (2021). <https://doi.org/10.1038/s41598-021-82017-6>
 26. Yazdani, E., Karamzadeh-Ziarati, N., Cheshmi, S.S., et al.: Automated segmentation of lesions and organs at risk on [68Ga]Ga-PSMA-11 PET/CT images using self-supervised learning with Swin UNETR. *Cancer Imaging* **24**(1), 30 (2024). <https://doi.org/10.1186/s40644-024-00675-x>
 27. Yousefirizi, F., Klyuzhin, I.S., O, J.H., et al.: TMTV-Net: Fully automated total metabolic tumor volume segmentation in lymphoma PET/CT images — a multi-center generalizability analysis. *Eur J Nucl Med Mol Imaging* **51**(7), 1937–1954 (2024). <https://doi.org/10.1007/s00259-024-06616-x>